

Daniyal Kashif

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EDUCATION

University of Arizona

Tucson, Arizona

Bachelor of Science, Computer Science - Minor in AI, Minor in Game Development

Expected May 2027

Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Computer Organization, Software Development, Linear Algebra, Principles of Data Science, Computational Linguistics, Systems Programming and Unix

Skills

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, C, SQL

ML/Data: Scikit-learn, XGBoost, PyTorch, Pandas, NumPy, NLTK, Streamlit, FastAPI

Web & Tools: Next.js, React, Django, PostgreSQL, Redis, Prisma, Vite, Clerk, Git, Figma

Work Experience

FBM Solutions

Tucson, USA

AI/ML Intern

June 2025-Aug 2025

- Developed and deployed an intelligent chatbot system using FastAPI, integrating document retrieval (RAG) with external API data sources.
- Fine-tuned DistilBERT transformer model for intent classification, achieving multi-class prediction across policy and data retrieval intents.
- Integrated external REST APIs with JWT authentication for real-time employee data retrieval and leave balance queries.

Auxin Security

Baltimore, Maryland

Security Research & Development Intern

June 2026-August 2026

- Researched and engineered a Zero Trust Network Exposure Function (NEF) gateway within a Service-Based Architecture, creating a fully functional proof-of-concept demonstration showcased at DEFCON.
- Secured API endpoints by integrating OAuth2 and JWT introspection, effectively mitigating Insecure Direct Object Reference (IDOR) and Server-Side Request Forgery (SSRF) vulnerabilities within the architecture.
- Developed and evaluated an AI-driven detection agent harness, utilizing industry research to analyze software development lifecycles and implement behavioral design models.

University of Arizona

Tucson, Arizona

Research Assistant

November 2023-September 2024

- Annotated and evaluated 500+ paragraphs using Human Signal and Label Studio, identifying key discrepancies between human and AI language understanding to support experimental NLP research.
- Collaborated with NLP researchers to compare differences in human vs. AI annotations across multi-subject sentence interpretation tasks.
- Investigated key limitations of large language models in parsing multi-verb or multi-subject inputs to support future model improvements.

Technical Projects

Watchlist TV Recommender | [source code](#) | Next.js · React · Python · Prisma · Tailwind CSS

- Built full-stack TV recommendation app with personalised user experiences using Next.js, React, and Python.
- Integrated Prisma ORM with TMDB API to fetch and manage show data across personalised recommendation pages.
- Built dynamic, reusable React components and implemented routing across 5+ pages in a collaborative team environment.

Network Intrusion Anomaly Detector | [source code](#) | Python · PyTorch · Scikit-learn · Streamlit

- Built unsupervised anomaly detection system comparing Isolation Forest vs PyTorch Autoencoder on 125K network connections from the NSL-KDD benchmark.
- Trained exclusively on normal traffic — model detects novel attack types it has never seen, achieving 0.93 F1 and 0.92 recall on the attack class.
- Tuned reconstruction error threshold to optimise F1, improving recall by 11% over the classical baseline.

Online Sexism Detector | [source code](#) | Python · XGBoost · TF-IDF · SVM · Streamlit

- Built NLP classifier achieving 0.81 weighted F1 on the EDOS benchmark dataset across 4,193 labelled examples.
- Benchmarked 6 model/feature combinations (Logistic Regression, SVM, XGBoost × Word/Char/Combined TF-IDF) to identify optimal pipeline.
- Combined word-level and character-level TF-IDF to capture both semantic patterns and misspelling/slang signals common in online toxic language.

Publications

- Kashif, D., [et al.]. "BEMEAE: Moving Beyond Exact Span Match for Event Argument Extraction." In **Proceedings of the NAACL 2025 Main Conference**. ACL Anthology, 2025. <https://aclanthology.org/2025.naacl-long.295/>